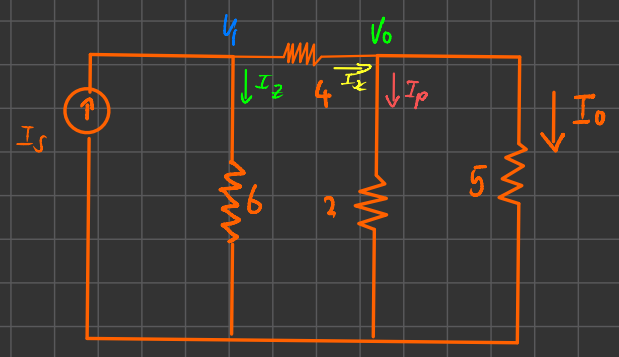
Linearity (assumptions)



$\frac{I_s}{I_o}$ (assume a value for one of them)

Assume $V_o = 5 \times I_o = 5V$

$I_o = 2A$ $I_p = \frac{V_o}{2} = \frac{5}{2} A$

I_n = out

KCL

$I_n = I_p + I_o = \frac{5}{2} + 1 = \frac{7}{2}$

V_1 } I_s unknow ∴ no no nodal

$\frac{V_1 - V_o}{4} = I_n$ (a form of nodal)

$\frac{V_1 - 5}{4} = \frac{7}{2}$

$V_1 = 19$

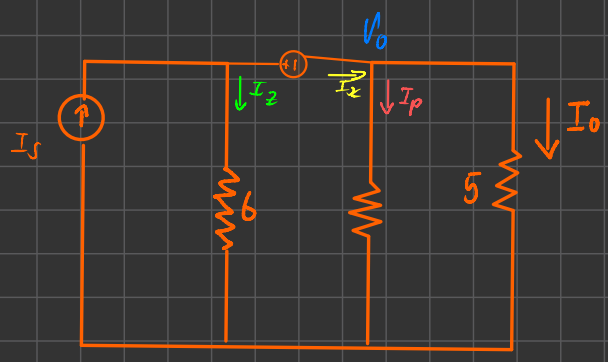
$I_2 = \frac{V_1}{6} = \frac{19}{6}$

KCL

$I_s = I_2 + I_n = \frac{19}{6} + \frac{7}{2}$

$I_s = 6,67$

$\frac{I_s}{I_o} = \frac{6,67}{1} = 6,67$



$\frac{I_o}{I_s}$ Assume $I_o = 2A$

$V_o = 5 \times 2 = 10$

$I_p = \frac{10}{5}$

$V_1 - V_o = 10$ (1)

$V_1 = 10 + 10 = 20$

Node 1 + Node 2 = 0

$\frac{V_1}{6} - I_s + \frac{V_o}{5} + \frac{V_o}{5} = 0$ (2)

$\frac{20}{6} - I_s + \frac{10}{5} + \frac{10}{5} = 0$

$I_s = \ldots$ $\frac{I_o}{I_s} = 2$

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